Mikhail Kornev

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RESEARCH AREAS

Algebraic Topology, Homotopy Theory, Group Theory, and Combinatorics

APPOINTMENTS

Sep. 2022 –	PhD student and research assistant at Steklov Mathematical Institute of RAS Department of Geometry and Topology
Mar. 2026 –	at HSE University International Laboratory for Mirror Symmetry and Automorphic Forms
Sep. 2021 – 2025	Research assistant at HSE University Laboratory of Algebraic Topology and Applications

EDUCATION

Sep. 2016 – May 2022	Specialist degree Diploma with honors in Mathematics (equivalent to M.S.) Lomonosov Moscow State University

RESEARCH PAPERS

Mathematics

Jan. 2026	*n-Valued Monoids on $\mathbb{C}P^1$ and Discriminants* joint with Victor Buchstaber Uspekhi Mat. Nauk, **81**:1(487) (2026), 207–208 https://doi.org/10.4213/rm10280
Jan. 2026	*On the Classification of n-Valued Monoids and Groups of Order* 3 Mat. Zametki, **119**:2 (2026), 244–252 https://doi.org/10.4213/mzm14803
Jan. 2024	*n-Valued Coset Groups and Dynamics* Mathematical Notes, **116**:1 (2024), 66–76, 2024 https://doi.org/10.4213/mzm14243
Sep. 2023	*Homology Spheres* Transactions of the Moscow Mathematical Society, **84** (2023), 197–239 https://doi.org/10.1090/mosc/352

Mathematics and Music

Sep. 2025	*The Technique of Musical Icosahedra* Music Academy, no. 3, 2025, pp. 64–83 https://doi.org/10.34690/479

PREPRINTS

Mar. 2026	Buchstaber, Ochanine, Krichever, and Witten genera `arXiv:2603.21118`
Oct. 2025	*Algebraic n-Valued Monoids on $\mathbb{C}P^1$, Discriminants and Projective Duality* joint with Victor Buchstaber `arXiv:2510.14010`
Aug. 2025	*On the Classification of n-Valued Monoids and Groups of Order* 3 `arXiv:2508.04454`
May 2025	*n-Valued Groups, Kronecker Sums, and Wendt's* (x, y, z)*-Matrices* joint with Victor Buchstaber `arXiv:2505.04296`
Jan 2024	n-valued Coset Groups and Dynamics `arXiv:2401.12718`

CONFERENCE TALKS

Aug. 2025	*n-Valued Groups, Kronecker Sums, and Wendt Matrices* joint work with Victor Buchstaber 5th Conference of Mathematical Centers of Russia
May 2024	*Homology Spheres, Acyclic Groups and Kan-Thurston Theorem* Conference "From Analysis to Homotopy Theory" , University of Greifswald, Greifswald, Germany
Apr. 2023	*Dynamics and multivalued groups* Conference "SIMC Youth Race", Steklov Mathematical Institute of RAS, Moscow, Russia
Oct. 2022	*Inconcretness of homotopy categories* International School "Toric topology, combinatorics and data analysis", Euler International Mathematical Institute, Saint Petersburg, Russia
Apr. 2022	*Acyclic groups and spaces* 29th International Scientific Conference for Undergraduate and Graduate Students and Young Scientists "Lomonosov", Moscow, Russia
Nov. 2021	*Acyclic groups and Kan-Thurston theorem*, International School "Toric Topology and Combinatoric", Sirius Mathematics Center, Sochi, Russia

SEMINAR TALKS

Dec. 2025	*Fermat hypersurfaces, projective duality, and algebraic n-valued monoids* Seminar of the Department of Geometry and Topology "Geometry, Topology and Mathematical Physics", Steklov Mathematical Institute of RAS (Novikov Seminar), Moscow, Russia
Dec. 2025	*Discrete and Algebraic n-Valued Groups* Seminar on Discrete Geometry and Geometry of Numbers, Moscow State University
Nov. 2025	*n-Valued Algebraic Monoids, Cubics, and Discriminants* The Seminar of International Laboratory of Algebraic Topology and Its Applications, HSE University, Moscow, Russia

May 2025	*n-Valued Groups, Kronecker Sums, and Wendt Matrices* Joint work with Victor Buchstaber Seminar of the Department of Geometry and Topology "Geometry, Topology and Mathematical Physics", Steklov Mathematical Institute of RAS (Novikov Seminar), Moscow, Russia
Apr. 2023	*Acyclic groups and Kan-Thurston theorem* Seminar by I. G. Lysenok and A. L. Talambutsa "Seminar on geometric group theory", Steklov Mathematical Institute of RAS, Moscow, Russia
Dec. 2022	*Fundamental groups of homology spheres* Seminar "Geometry, Topology and their applications" at Sobolev Institute of Mathematics

GRANTS AND AWARDS

Prizes, fellowships and grants

2021 –	*The Russian Science Foundation Grant No. 23-11-00143*
2019	*Winner of the olympiad on Partial Differential Equations* Moscow State University
2018	*Winner of the olympiad on Ordinary Differential Equations* Moscow State University

STUDY SEMINAR TALKS

Texas Tech Quantum Homotopy Seminar

Dec. 2024	*An Introduction to Cartan Geometries*
Mar. 2024	*Euler-Lagrange dynamics via the ∞-jet bundle* (2 talks)

Seminar On Homotopy Theory and Applications at Independent University of Moscow

Dec. 2023	*Principal ∞-bundles*
Apr. – May 2023	*A mini-course on the ∞-category of motivic coarse spaces* (4 talks)
Apr. 2023	*Elements of unstable motivic homotopy theory*
Feb. – Apr. 2023	*A mini-course on ∞-cosmoi* (4 talks)
Dec. 2022	*Applications of Reedy theory*
Nov. 2022	*Weighted (co-)limits, Gambino's theorem, Reedy model structures*
Nov. 2022	*Homotopically final functors and examples of calculations with homotopy (co-)limits*
Nov. 2022	*Nerve—realization adjunction and bar construction*
Apr. 2022	*A series of 4 talks on sheaves of sets*
Apr. 2022	*Kan extensions. Additional topics*
Oct. 2021	*The categorical meaning of the Kan-Thurston theorem*
Jul. – Aug. 2021	*Quillen's theorem, a closed model structure on sSet*

Study And Scientific Seminar "Geometry And Topology" at Moscow State University

Sep. 2020	*Andreev's theorem on hyperbolic polyhedra*
Nov. 2019	*The intersection form of 4-manifolds*
May 2019	*Nilmanifolds and Massey product in bigraded cohomology rings*

CONFERENCES, WORKSHOPS AND SCHOOLS

May 2024	*Conference "From Analysis to Homotopy Theory"* University of Greifswald, Greifswald, Germany
Jul. 2023	*Summer School "Algebra and Geometry"* HSE University, Steklov Mathematical Institute of RAS, Suzdal, Russia
Jul. 2023	*Summer school on non-commutative and derived geometry of matrix factorisations* Centre of Pure Mathematics, Moscow Institute of Physics and Technology, Dolgoprudny, Russia
Jun. 2023	*Conference on algebra, algebraic geometry, and number theory on the occasion of academician Igor Rostislavovich Shafarevich 100th birthday* Steklov Mathematical Institute of RAS, Moscow, Russia
Apr. 2023	*Conference "Geometry, topology and mathematical physics" on the occasion of the 85th anniversary of S. P. Novikov and the 80th anniversary of V. M. Buchstaber* Steklov Mathematical Institute of RAS, Moscow, Russia
Mar. 2023	*Conference "SIMC Youth Race"* Steklov Mathematical Institute of RAS, Moscow
Oct. 2022	*International School "Toric topology, combinatorics and data analysis"* Euler International Mathematical Institute, Saint Petersburg, Russia
Jul. 2022	*Summer School "Algebra and Geometry"* HSE University, Steklov Mathematical Institute of RAS, Suzdal, Russia
Apr. 2022	*29th International Scientific Conference for Undergraduate and Graduate Students and Young Scientists "Lomonosov"* Lomonosov Moscow State University, Moscow, Russia
Nov. 2021	*International School "Toric Topology and Combinatorics"* HSE University, Sirius Mathematics Center, Sochi, Russia
Nov. 2020	*Topology and Geometry of Group Actions* HSE University, online
Nov. 2019	7th Summer School On Geometrical Methods Of Mathematical Physiques Landau Institute for Theoretical Physics, Lomonosov Moscow State University, Chernogolovka, Russia
Nov. 2019	*Geometry, Topology And Mathematical Physics* Landau Institute for Theoretical Physics, Lomonosov Moscow State University, Chernogolovka, Russia
Nov. 2018	*Geometry, Topology And Mathematical Physics* Landau Institute for Theoretical Physics, Lomonosov Moscow State University, Chernogolovka, Russia
Jul. 2018	*Summer school "Modern Mathematics"* HSE University, Lomonosov Moscow State University, Independent University of Moscow, Dubna, Russia

Jul. 2017	*Summer school "Modern Mathematics"* HSE University, Lomonosov Moscow State University, Independent University of Moscow, Dubna, Russia

TEACHING

May 2022 –	*Teaching various courses (e. g., Algebraic Geometry, Algebraic Topology, Abstract Algebra, Differential Geometry, Linear Algebra, Probability Theory, Discrete Mathematics, Calculus, etc.) on Profi* `https://profi.ru/profile/KornevMI3/`
Sep. 2016 – 2022	*Little Mech-Math courses* An olympiad Math class at Lomonosov Moscow State University `http://mmmf.msu.ru/about/person.html`

SERVICE

June 2021 – 2024	*Co-organization of the study seminar "Homotopy Theory and Applications" at Independent University of Moscow* `https://researchseminars.org/seminar/HomotopyTheoryAndApps`

REFERENCES

Victor Buchstaber, Steklov Mathematical Institute of RAS
`buchstab@mi-ras.ru`

LANGUAGES

Russian (native)

English (fluent)

French (intermediate)

Latin (basic)